

Homework #3:

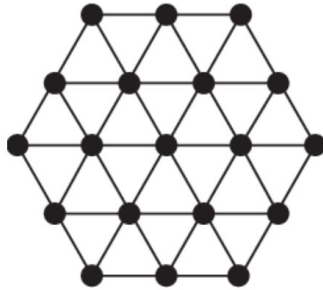
Due date: Oct. 19th 2025

Crystallographic restriction theorem

1. Proof the crystallographic restriction theorem: Consider a crystal (i.e. with translational symmetry) in three dimensions that is invariant under rotations by an angle $2\pi/n$ around an axis. Then the only allowed possibilities are $n = 1, 2, 3, 4, 6$ with the case $n = 1$ being the trivial case with no rotational symmetry.

2D triangular lattice

2. Let us consider a two-dimensional triangular lattice as shown in following Figure.
 - (a) Identify the primitive unit cell of the two-dimensional triangular lattice. Find the basis vectors.
 - (b) Construct the basis vectors of the reciprocal unit cell.



Properties of the reciprocal lattice

3.
 - (a) Show that all reciprocal lattice vectors of the form $\mathbf{G} = h\mathbf{A} + k\mathbf{B} + l\mathbf{C}$ are perpendicular to the planes of the same indices (h, k, l) in real space.
 - (b) Show that the distance $d_{h,k,l}$ between two consecutive planes (h, k, l) is inversely proportional to $|\mathbf{G}_{h,k,l}|$.
 - (c) Find $d_{h,k,l}$ for a simple cubic lattice and an orthorhombic lattice ($a \neq b \neq c, \alpha = \beta = \gamma = \pi/2$).

Atomic planes and Miller indices: application to lithium

4. The Bravais lattice of lithium is simple cubic with lattice parameter $a = 3.48 \text{ \AA}$.
 - (a) Suppose that the atoms (assumed to be spheres) are placed along the $[111]$, represent the atomic distribution along the following faces (100) , (110) , (111) , and (201) .
 - (c) For each such 2D structure, state the direction and basis vector $\mathbf{a}$ and $\mathbf{b}$ of the elementary lattice as well as the value of the angle, γ .
 - (d) Find the atomic concentration and the mass density of lithium ($A \approx 7$).